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Exploring Periodicity in Economic Data

For my visualization project 3, I plan on exploring ways to visualize the periodic nature of some economic phenomenon. Specifically, I plan on designing and building a novel visualization for CPI data and testing it on data from the United States. The first part of the visualization will be to construct a normal line chart based on the CPI data and the timestep. The second, novel part of the visualization will be to construct a circular glyph, with each level corresponding to, for example, one year's worth of data. In other words, there will be a circle that encompasses all the displayed data and, every x distance on the radius, there will be some increment of displayed CPI data. Most likely, this data will be displayed like a line chart, with its base/zero as the circumference of the radius at the top of the previous data.

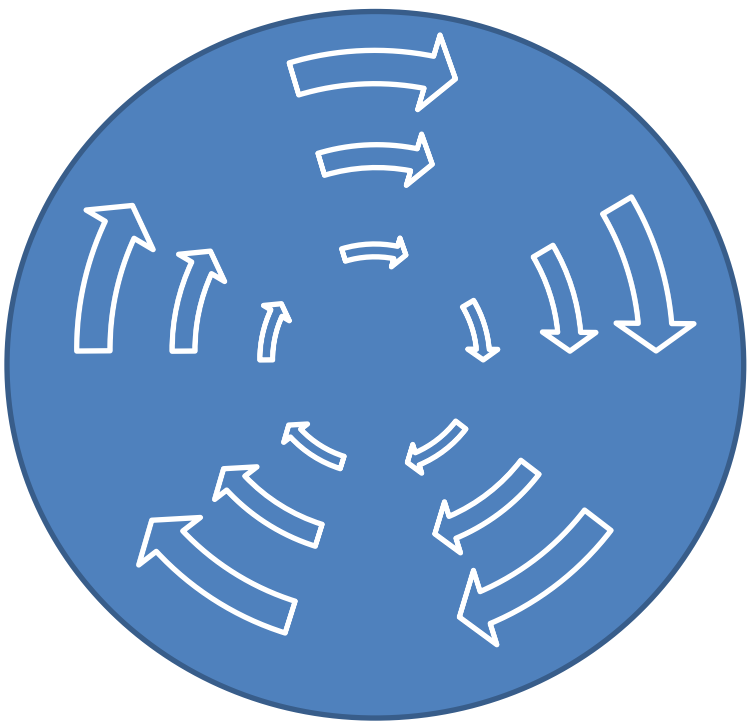

Figure 1 A mock-up of the proposed visualization, where the arrows represent the data of a single displayed increment. Note that in the actual visualization each set of arrows will be replaced by a line graph,

The primary goal for this visualization is to illustrate the periodicity of the data. Ideally, using this visualization will allow users to find less noticeable patterns in the data than they would be able to find without using this visualization. This visualization will also allow users to view the data normally, if they wish to look more specifically at exact numbers.

To create this visualization, I will most likely use D3.js, a javascript library, to produce this visualization and display it on the web. I plan on including a normal line chart and buttons for selecting the amount of information to display, as well as switching between the normal line chart and the circle visualization. I may also include some buttons to allow users to control how many levels the circular visualization has.

In my search through the visualization literature, I was not able to find many examples of similar visualizations. One example I found that dealt with visualizing CPI data took a different, more simplistic approach [1]. It simply graphed each month as a bubble (without labeling the months, as far as I can tell) and then plotted the ten-year inflation rate vs. 2012's inflation rate. I found a couple of other papers that dealt with visualizing economic data, but, in both, most of the elements were unrelated. I know, from a previous visualization class, that circular graphs similar to what I propose have been previously used in visualization research, particularly in illustrating periodic trends. In one example that I was able to find, one group of researchers created a spiral graph to illustrate the patterns of sunlight and darkness over the course of several days [2]. This visualization, while notably different than my proposed visualization, strongly resembles what I am attempting here. It uses more or less the same type of display that I am using; however, instead of using line graphs to represent the sunlight, this visualization uses color.

Producing this visualization may create some unusual challenges. For example, my current plan is to split up the circle by giving each line graph an equal length of the radius. In other words, for each line graph, the y-axis will be the same size. However, due to the nature of circles, this means that as the line graphs move closer to the outside of the circle, the x-axis will grow noticeably longer. I do not expect for this to be a problem in the visualization, but it could cause some unexpected side effects.

Also, I am unfamiliar with the visualization field's methods of testing the effectiveness of a visualization. Likely, these methods would require some significant testing with human subjects, which would, of course, be outside the bounds of this class. Instead, I plan on manually comparing the circle visualization with the normal line charts to see which better illustrates periodicity. I suspect that the circle graph will be a better indicator of periodicity than the line graph, but that the line graph will be better at showing the specifics of the data. Either way, it will be interesting to see how well the circle visualization performs.

References

[1] B. Meyer, “Visualizing Disinflation…And No, We’re Not There Yet,” Inflation and Price Statistics.

[2] M. Alexa, W. Müller, M. Weber, “Visualizing Time-Series on Spirals.”